Computer Analog

The Future of Computing

Activity Book

Name Keshav Narayan

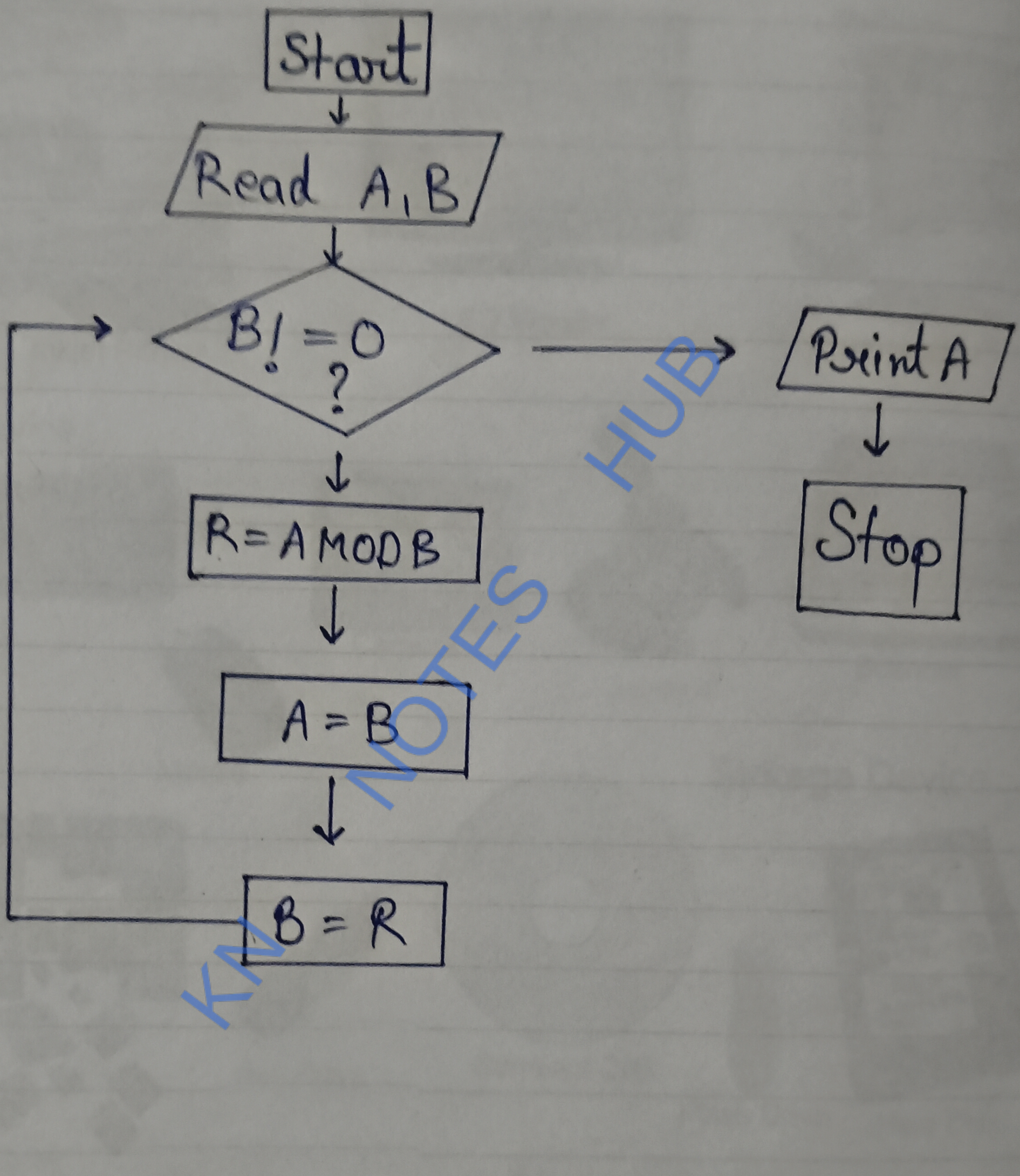
Class B.Tech (AIML) Sec. _____

Subject C.S.E

School Deen Dayal Upadhaya University

KN NOTES HUB

Flowchart →



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Assignment → 01

Algorithms and flowcharts

Ques: Create an algorithm and flowchart that will output the Greatest Common factor (GCf) of two numbers?

Sol→ **Algorithm →**

1. Start the program
2. Input two positive A and B.
3. Find the remainder R = A MOD B
4. If R = 0, then B is the GCf. Go to step (7).
5. Otherwise, Set A = B and B = R, the repeat the step 3.
6. Continue until the remainder become zero.
7. Output B as the Gr. C. f.
8. Stop the program.

→ Example →

Given A = 48, B = 18

48 MOD 18 = 12 → A = 18, B = 12

18 MOD 12 = 6 → A = 12, B = 6

12 MOD 6 = 0 → Gr. C. f = 6

Output: GCf = 6

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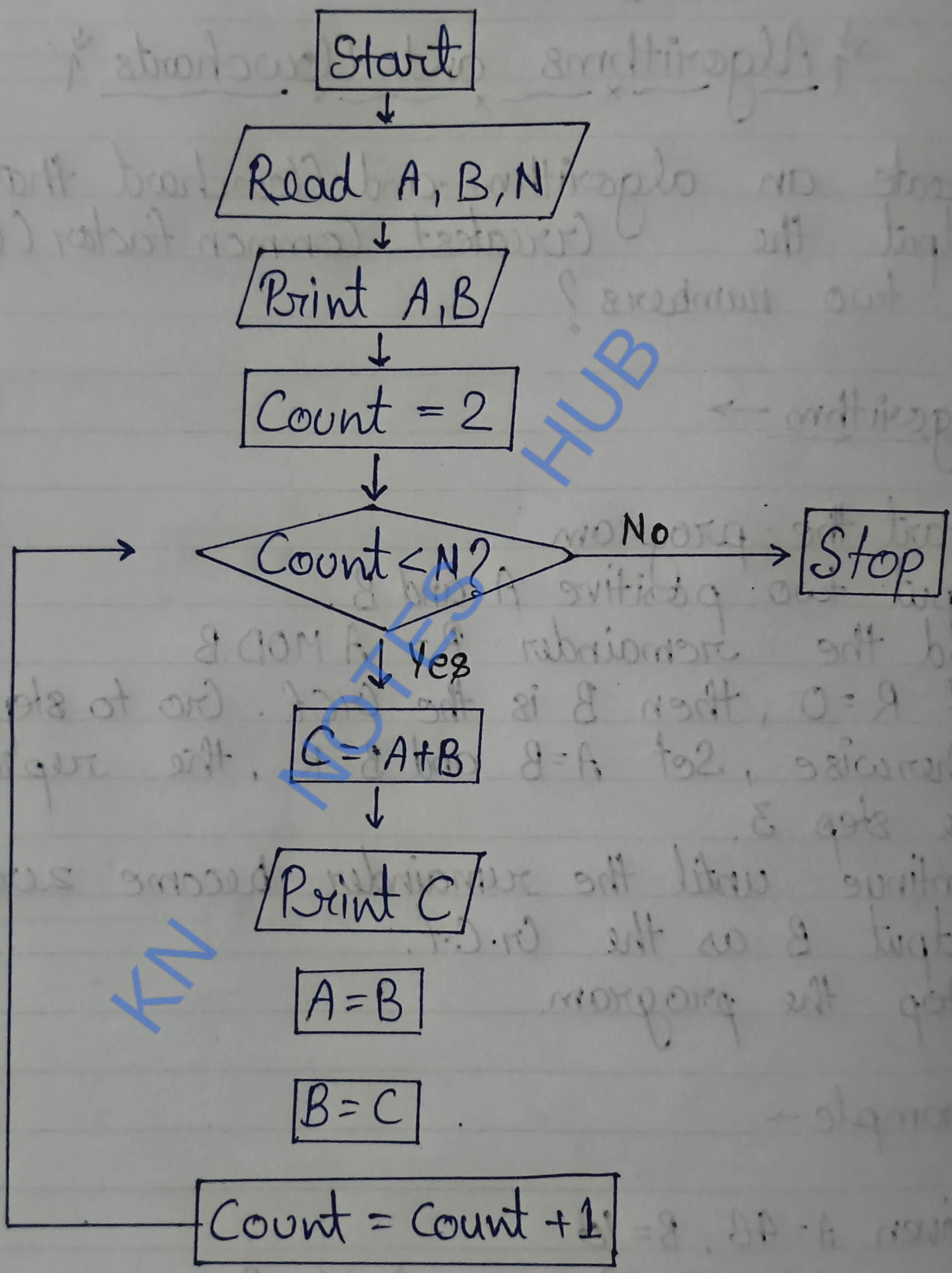
Flowchart →



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Ques (2) Create an algorithm and flowchart that will output the fibonacci series for two given numbers

Sol- Algorithm →

1. Start the program
2. Input two numbers LOW and HIGH
3. Set A=0 and B=1
4. Repeat while A ≤ HIGH
5. If A ≥ LOW, Print A
6. Calculate C= A+B
7. Set A = B and B = C
8. Repeat Steps 4-7 untill A becomes
9. Greater than HIGH
10. Stop the program

Example →

Given: LOW = 5 HIGH = 50

fibanacci sequence 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55 ....

Terms between 5 and 50 are 5, 8, 13, 21, 34

Output : 5, 8, 13, 21, 34

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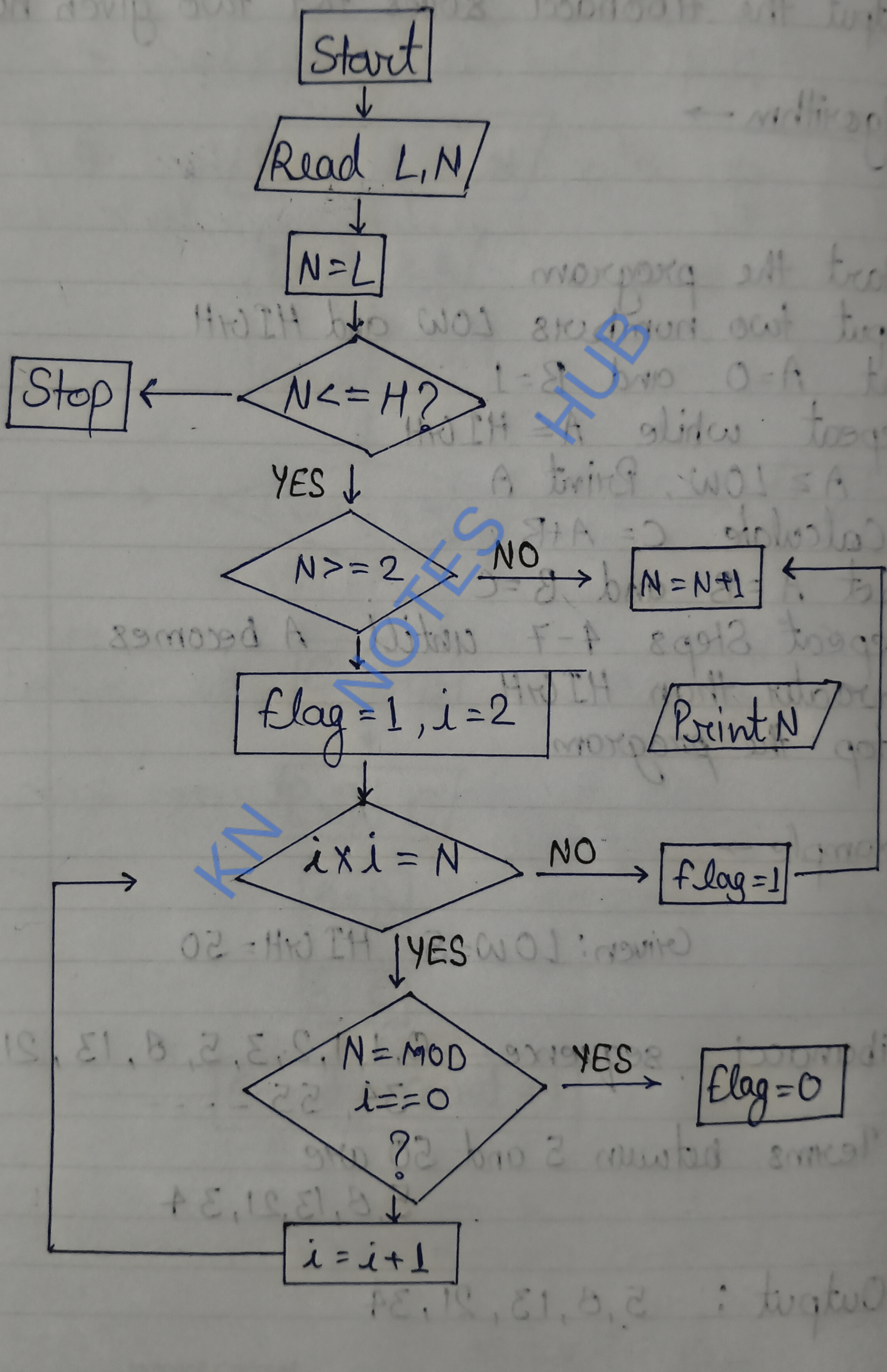
Flowchart →



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Ques (3) Create an algorithm and flowchart that will output all prime numbers between two given numbers?

Algorithm →

1. Start the program
2. Input the number LOW and HIGH
3. Set "N = LOW"
4. Check each number "N" while N ≤ HIGH
5. If N is less than 2, move to the next number.
6. Set I = 2 and assume N = Prime
7. Check divisiblity of (N) by (I) while → I × I ≤ N
8. If N = MOD I = 0, N is not prime number
9. If no divisor is found, output (N) as prime number.
10. Increase (N) by 1 and repeat the checking number process.
11. Stop when (N) become greater than HIGH

* Example → Given: Low = 10, High = 30

Prime number between 10 and 30 are: 11, 13, 17, 19, 23, 29

Output: 11, 13, 17, 19, 23, 29

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